

KCET – Chemistry – 2019  
VERSION CODE: C-1

**Question 1:** Which of the following possess net dipole moment?

- (A)  $\text{SO}_2$
- (B)  $\text{BF}_3$
- (C)  $\text{BeCl}_2$
- (D)  $\text{CO}_2$

**Answer 1: (A)**

**Question 2:** The number of  $\pi$ -bonds and  $\sigma$ -bonds present in naphthalene are respectively

- (A) 6, 19
- (B) 5, 19
- (C) 5, 11
- (D) 5, 20

**Answer 2: (B)**

**Question 3:** The reaction in which  $\Delta H > \Delta U$  is

- (A)  $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{NO}(\text{g})$
- (B)  $\text{CaCO}_3(\text{s}) \rightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$
- (C)  $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$
- (D)  $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$

**Answer 3: (B)**

**Question 4:** The number of moles of electron required to reduce 0.2 mole of  $\text{Cr}_2\text{O}_7^{2-}$  to  $\text{Cr}^{+3}$

- (A) 1.2
- (B) 6
- (C) 12
- (D) 0.6

**Answer 4: (A)**

**Question 5:** In the reaction  $\text{B}(\text{OH})_3 + 2\text{H}_2\text{O} \rightarrow [\text{B}(\text{OH})_4]^- + \text{H}_3\text{O}^+$   $\text{B}(\text{OH})_3$  functions are

- (A) Protonic acid
- (B) Lewis base
- (C) Bronsted acid
- (D) Lewis acid

**Answer 5: (D)**

**Question 6:** Match the following acids with their  $\text{pK}_a$  values:

| Acid |               | pKa  |      |
|------|---------------|------|------|
| a.   | Phenol        | i.   | 16   |
| b.   | p-Nitrophenol | ii.  | 0.78 |
| c.   | Ethanol       | iii. | 10   |
| d.   | Picric acid   | iv.  | 7.1  |

|     | a   | b  | c   | d   |
|-----|-----|----|-----|-----|
| (A) | iii | iv | i   | iii |
| (B) | ii  | i  | ii  | iv  |
| (C) | iii | i  | iv  | ii  |
| (D) | iv  | ii | iii | i   |

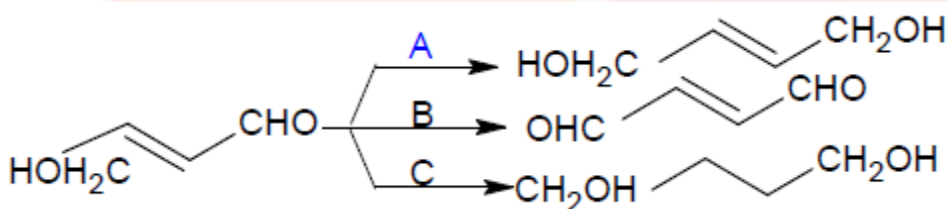
**Answer 6: (A)**

**Question 7:** Which of the following can be used to test the acidic nature of ethanol?

- (A) Blue litmus
- (B)  $\text{Na}_2\text{CO}_3$
- (C)  $\text{NaHCO}_3$
- (D) Na metal

**Answer 7: (D)**

**Question 8:**



The reagents A, B and C respectively are  $\text{CH}_2\text{OH}$

- (A)  $\text{H}_2/\text{Pd}$ , PCC,  $\text{NaBH}_4$
- (B)  $\text{NaBH}_4$ , alk,  $\text{KMnO}_4$ ,  $\text{H}_2/\text{Pd}$
- (C)  $\text{NaBH}_4$ , PCC,  $\text{H}_2/\text{Pd}$
- (D)  $\text{H}_2/\text{Pd}$ , alk,  $\text{KMnO}_4$ ,  $\text{NaBH}_4$ .

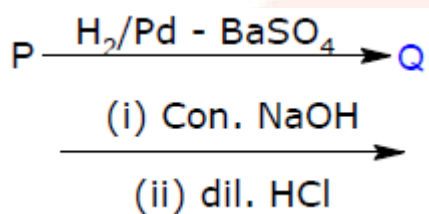
**Answer 8: (C)**

**Question 9:** Propanoic acid undergoes HVZ reaction to give chloropropanoic acid. The product obtained is

- (A) Stronger acid than propanoic acid
- (B) As stronger as propanoic acid
- (C) Weaker acid than propanoic acid
- (D) Stronger than dichloropropanoic acid

**Answer 9: (A)**

**Question 10:**



R and S form benzyl benzoate when treated with each other. Hence P is

- (A)  $\text{C}_6\text{H}_5\text{CHO}$
- (B)  $\text{C}_6\text{H}_5\text{CH}_2\text{OH}$
- (C)  $\text{C}_6\text{H}_5\text{COCl}$
- (D)  $\text{C}_6\text{H}_5\text{COOH}$

**Answer 10: (C)**

**Question 11:** Among the following the main reactions occurring in blast furnace during extraction of iron from haematite are

- (i)  $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$
- (ii)  $\text{FeO} + \text{SiO}_2 \rightarrow \text{FeSiO}_3$
- (iii)  $\text{Fe}_2\text{O}_3 + 3\text{C} \rightarrow 2\text{Fe} + 3\text{CO}$
- (iv)  $\text{CaO} + \text{SiO}_2 \rightarrow \text{CaSiO}_3$

- (A) (i) and (ii)
- (B) (iii) and (iv)
- (C) (ii) and (iii)
- (D) (i) and (iv)

**Answer 11: (D)**

**Question 12:** Which of the following pair contains 2 lone pair of electrons on the central atom?

- (A)  $\text{I}_3^+$ ,  $\text{H}_2\text{O}$
- (B)  $\text{H}_2\text{O}$ ,  $\text{NF}_3$

- (C)  $\text{XeF}_4$ ,  $\text{NH}_3$   
(D)  $\text{SO}_4^{2-}$ ,  $\text{H}_2\text{S}$

**Answer 12: (A)**

**Question 13:** Which of the following statement is correct?

- (A)  $\text{Cl}_2$  oxidises  $\text{H}_2\text{O}$  to  $\text{O}_2$  but  $\text{F}_2$  does not.  
(B)  $\text{Cl}_2$  is a stronger oxidizing agent than  $\text{F}_2$ .  
(C)  $\text{F}_2$  oxidises  $\text{H}_2\text{O}$  to  $\text{O}_2$  but  $\text{Cl}_2$  does not  
(D) Fluoride is a good oxidising agent.

**Answer 13: (C)**

**Question 14:** 0.1 mole of  $\text{XeF}_6$  is treated with 1.8 g of water. The product obtained is

- (A)  $\text{XeO}_3$   
(B)  $\text{XeO}_2\text{F}_2$   
(C)  $\text{XeOF}_4$   
(D)  $\text{Xe} + \text{XeO}_3$

**Answer 14: (C)**

**Question 15:** In the reaction of gold with aquaregia, oxidation state of Nitrogen changes from

- (A) +4 to +2  
(B) +6 to +4  
(C) +5 to +2  
(D) +3 to +1

**Answer 15: (C)**

**Question 16:** The vitamin that helps to clotting of blood is

- (A) A  
(B) C  
(C)  $\text{B}_2$   
(D) K

**Answer 16: (D)**

**Question 17:** The polymer containing five methylene groups in its repeating unit is

- (A) Nylon 6, 6  
(B) Nylon 6  
(C) Dacron  
(D) Bakelite

**Answer 17: (B)**

**Question 18:** Cis – 1, 4 – polyisoprene is called

- (A) Buna – N
- (B) Neoprene
- (C) Buna-S
- (D) Natural rubber

**Answer 18: (D)**

**Question 19:** Which cleansing agent gets precipitated in hard water?

- (A) Sodium lauryl sulphate
- (B) Sodium stearate
- (C) Cetyl trimethyl ammonium bromide
- (D) Sodium dodecyl benzene sulphonate

**Answer 19: (B)**

**Question 20:** Anti-histamine among the following is

- (A) Bromopheneramine
- (B) Morphine
- (C) Amoxycillin
- (D) Chloroxylonol

**Answer 20: (A)**

**Question 21:** The elements in which electrons are progressively filled in 4f orbital are called

- (A) Actinoids
- (B) Transition elements
- (C) Lanthanoids
- (D) Halogens

**Answer 21: (C)**

**Question 22:** Incorrect statement with reference to Ce ( $Z = 58$ )

- (A)  $Ce^{4+}$  is a reducing agent.
- (B) Ce in +3 oxidation state is more stable than in +4.
- (C) Atomic size of Ce is more than that of Lu.
- (D) Ce shows common oxidation states of +3 and +4

**Answer 22: (A)**

**Question 23:** A mixture of  $NaC/$ and  $K_2Cr_2O_7$  is heated with conc.  $H_2SO_4$ , deep red vapours and formed. Which of the following statements is false?

- (A) The vapours give a yellow solution with NaOH.
- (B) The vapours contain  $CrO_2Cl_2$  only.
- (C) The vapours contain  $CrO_2Cl_2$  and  $Cl_2$ .
- (D) The vapours when passed into lead acetate in acetic give a yellow precipitate.

**Answer 23: (C)**

**Question 24:** Which of the following statements is wrong?

- (A) In highest oxidation states, the transition metals show acidic character.
- (B)  $\text{Mn}^{3+}$  and  $\text{CO}^{3+}$  are oxidizing agents in aqueous solution.
- (C) Metals in highest oxidation states are more stable in oxide than in fluorides.
- (D) All elements of 3d series exhibit variable oxidation states.

**Answer 24: (D)**

**Question 25:** Which among the following is the strongest ligand?

- (A)  $\text{CN}^-$
- (B)  $\text{NH}_3$
- (C)  $\text{CO}$
- (D) en

**Answer 25: (C)**

**Question 26:** Which of the following is a network crystalline solid?

- (A)  $\text{I}_2$
- (B)  $\text{AlN}$
- (C)  $\text{NaCl}$
- (D) Ice

**Answer 26: (B)**

**Question 27:** The number of atoms in 2.4 g of body centred cubic crystal with edge length 200 pm is

(density =  $10 \text{ g cm}^{-3}$ ,  $N_A = 6 \times 10^{23} \text{ atoms/mol}$ )

- (A)  $6 \times 10^{22}$
- (B)  $6 \times 10^{20}$
- (C)  $6 \times 10^{23}$
- (D)  $6 \times 10^{19}$

**Answer 27: (A)**

**Question 28:** 1mole of  $\text{NaCl}$  is doped with  $10^{-5}$  mole of  $\text{SrCl}_2$ . The number of cationic vacancies in the crystal lattice will be

- (A)  $6.022 \times 10^{18}$
- (B)  $6.022 \times 10^{15}$
- (C)  $6.022 \times 10^{23}$
- (D)  $12.044 \times 10^{20}$

**Answer 28: (A)**

**Question 29:** A non-volatile solute, 'A' tetramerises in water to the extent of 80%. 2.5g of 'A' in 100 g of water, lower the freezing point by  $0.3^\circ\text{C}$ . The molar mass of A in  $\text{mol L}^{-1}$  is ( $K_f$  for water =  $1.86 \text{ K kg mol}^{-1}$ )

(A) 62

(B) 221

(C) 155

(D) 354

**Answer 29: (A)**

**Question 30:** Solution 'A' contains acetone dissolved in chloroform and solution 'B' contains acetone dissolved in carbon disulphide. The type of deviations from Raoult's law shown by solutions A and B, respectively are

(A) Positive and positive

(B) Positive and negative

(C) Negative and negative

(D) Negative and positive

**Answer 30: (d)**

**Question 31:** The mass of AgCl precipitated when a solution containing 11.70 g of NaCl is added to a solution containing 3.4g of AgNO<sub>3</sub> is [Atomic mass of Ag = 108, Atomic mass of Na = 23]

(A) 5.74 g

(B) 1.17 g

(C) 2.87 g

(D) 6.8 g

**Answer 31: (C)**

**Question 32:** Two particle A and B are in motion. If the wavelength associated with 'A' is 33.33 nm, the wavelength associated with 'B' whose momentum is  $\frac{1}{3}$  rd of 'A' is

(A)  $1.0 \times 10^{-8}$ m

(B)  $2.5 \times 10^{-8}$ m

(C)  $1.25 \times 10^{-7}$  m

(D)  $1.0 \times 10^{-7}$ m

**Answer 32: (D)**

**Question 33:** The first ionization enthalpy of the following elements are in the order:

(A) C < N < Si < P

(B) P < Si < N < C

(C) P < Si < C < N

(D) Si < P < C < N

**Answer 33: (D)**

**Question 34:** Solubility of AgCl is least in

- (A) 0.1 M NaCl
- (B) Pure water
- (C) 0.1 M BaCl<sub>2</sub>
- (D) 0.1 M AlCl<sub>3</sub>

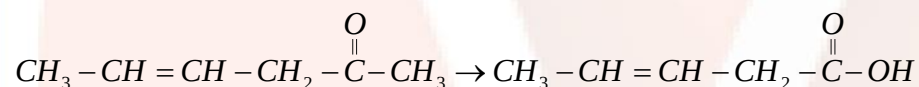
**Answer 34: (D)**

**Question 35:** Which of the following equations does NOT represent Charles's law for a given mass of gas at constant pressure?

- (A)  $\frac{V}{T} = K$
- (B)  $\log V = \log K + \log T$
- (C)  $\log K = \log V + \log T$
- (D)  $\frac{d(\ln V)}{dT} = \frac{1}{T}$

**Answer 35: (C)**

**Question 36:** Which is the most suitable reagent for the following conversion?



- (A) Tollen's reagent
- (B) I<sub>2</sub> and NaOH solution
- (C) Benzoyl peroxide
- (D) Sn and NaOH solution

**Answer 36: (B)**

**Question 37:** Which of the following is least soluble in water at 298K?

- (A) CH<sub>3</sub>NH<sub>2</sub>
- (B) (CH<sub>3</sub>)<sub>3</sub>N
- (C) (CH<sub>3</sub>)<sub>2</sub>NH
- (D) C<sub>6</sub>H<sub>5</sub>NH<sub>2</sub>

**Answer 37: (B)**

**Question 38:** If Aniline is treated with 1: 1 mixture of con. HNO<sub>3</sub> and con. H<sub>2</sub>SO<sub>4</sub>, p-nitroaniline and m-nitroaniline are formed nearly in equal amounts. This is due to

- (A) m-directing property of -NH<sub>2</sub> group
- (B) Protonation of -NH<sub>2</sub> which causes deactivation of benzene ring
- (C) m and p directing property of -NH<sub>2</sub> group
- (D) Isomerisation of some p-nitroaniline into m-nitroaniline

**Answer 38: (B)**

**Question 39:** In nucleic acids, the nucleotides are joined together by

- (A) Phosphoester linkage
- (B) Phosphodiester linkage
- (C) Phosphodisulphide linkage
- (D) Sulphodiester linkage

**Answer 39: (B)**

**Question 40:** Which of the following is generally water insoluble?

- (A) Fibrous protein
- (B) Vitamin-C
- (C) Amylose
- (D) Glycine

**Answer 40: (A)**

**Question 41:** Relative lowering of vapour pressure of a dilute solution of glucose dissolved in 1kg of water is 0.002. The molality of the solution is

- (A) 0.004
- (B) 0.222
- (C) 0.111
- (D) 0.021

**Answer 41: (C)**

**Question 42:** One litre solution of  $\text{MgCl}_2$  is electrolyzed completely by passing a current of 1A for 16 min 5 sec. The original concentration of  $\text{MgCl}_2$  solution was (Atomic mass of Mg = 24)

- (A)  $5 \times 10^{-3}\text{M}$
- (B)  $5 \times 10^{-2}\text{M}$
- (C)  $0.5 \times 10^{-3}\text{M}$
- (D)  $1.0 \times 10^{-2}\text{M}$

**Answer 42: (A)**

**Question 43:** An aqueous solution of  $\text{CuSO}_4$  is subjected to electrolysis using inert electrodes. The pH of the solution will

- (A) increase
- (B) remains unchanged
- (C) decrease
- (D) increase or decrease depending on the strength of the current

**Answer 43: (C)**

**Question 44:** Give  $E^0_{\text{Mn}^{+7}|\text{Mn}^{+2}} = 15\text{V}$  and  $E^0_{\text{Mn}^{+4}|\text{Mn}^{+2}} = 1.2\text{V}$  then,  $E^0_{\text{Mn}^{+7}|\text{Mn}^{+4}}$  is

(A) 0.3V

(B) 0.1 V

(C) 1.7 V

(D) 2.1 V

**Answer 44 : (C)**

**Question 45:** The plot of  $t_{1/2}$  v/s  $[R]_0$  for a reaction is a straight-line parallel to x-axis. The unit for the rate constant of this reaction is

(A)  $\text{mol L}^{-1}\text{s}$

(B)  $\text{mol L}^{-1}\text{s}^{-1}$

(C)  $\text{L mol}^{-1} \text{s}^{-1}$

(D)  $\text{s}^{-1}$

**Answer 45: (D)**

**Question 46:** The metal nitrate that liberates  $\text{NO}_2$  on heating

(A)  $\text{NaNO}_3$

(B)  $\text{LiNO}_3$

(C)  $\text{KNO}_3$

(D)  $\text{RbNO}_3$

**Answer 46: (B)**

**Question 47:** Which of the following is NOT true regarding the usage of hydrogen as a fuel?

(A) High calorific value

(B) The combustible energy of hydrogen can be directly converted to electrical energy in a fuel cell

(C) Combustion product is ecofriendly.

(D) Hydrogen gas can be easily liquefied and stored.

**Answer 47: (D)**

**Question 48:** Resonance effect is not observed in

(A)  $\text{CH}_2 = \text{CH} - \text{CH} = \text{CH}_2$

(B)  $\text{CH}_2 = \text{CH} - \text{C} \equiv \text{N}$

(C)  $\text{CH}_2 = \text{CH} - \text{Cl}$

(D)  $\text{CH}_2 = \text{CH} - \text{CH}_2 - \text{NH}_2$

**Answer 48: (D)**

**Question 49:** 2-butyne is reduced to trans-but-2-ene using

(A)  $\text{H}_2/\text{Ni}$

(B) Na in liq.  $\text{NH}_3$

- (C)  $\text{H}_2$  | Pd-C  
(D) Zn in dil. HCl

**Answer 49: (B)**

**Question 50:** Eutrophication causes

- (A) Increase of nutrients in water  
(B) Reduction in water pollution  
(C) Reduction in dissolved oxygen  
(D) Decreases BOD

**Answer 50: (C)**

**Question 51:** Addition of excess of  $\text{AgNO}_3$  to an aqueous solution of 1 mole of  $\text{PdCl}_2 \cdot 4\text{NH}_3$  gives 2 moles of AgCl. The conductivity of this solution corresponds to

- (A) 1:1 electrolyte  
(B) 1:3 electrolyte  
(C) 1:2 electrolyte  
(D) 1:4 electrolyte

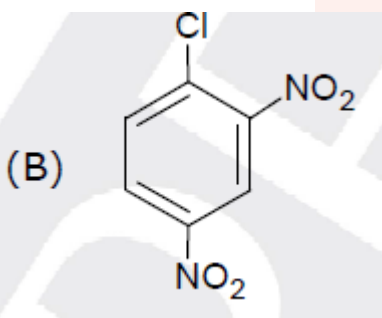
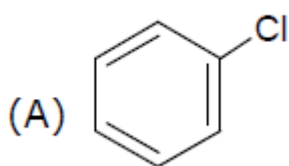
**Answer 51: (C)**

**Question 52:** The formula of penta aquanitrato chromium (III) nitrate is,

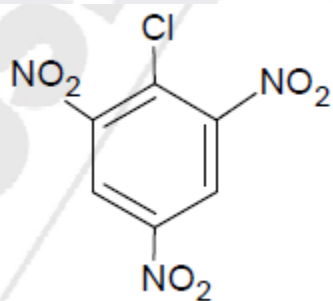
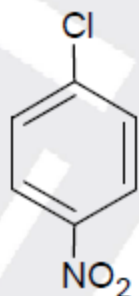
- (A)  $[\text{Cr}(\text{H}_2\text{O})_6(\text{NO}_3)_3]$   
(B)  $[\text{Cr}(\text{H}_2\text{O})_6(\text{NO}_2)_2]$   
(C)  $[\text{Cr}(\text{H}_2\text{O})_5\text{NO}_3](\text{NO}_3)_2$   
(D)  $[\text{Cr}(\text{H}_2\text{O})_5\text{NO}_2]\text{NO}_3$

**Answer 52: (C)**

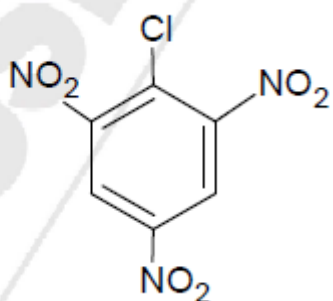
**Question 53:** Which of the following halide undergoes hydrolysis on warming with water/aqueous NaOH?



(C)



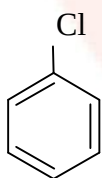
(D)



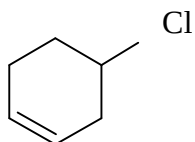
**Answer 53: (D)**

**Question 54:** The compound having longest C- Cl bond is

(A)

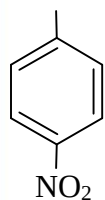


(B)



(C)

Cl



(D) CH<sub>2</sub> = CH – Cl

**Answer 54: (B)**

**Question 55:** The alkyl halides required to prepare by Wurtz reaction are

(A) Cl and Cl

(B) and Cl

(C) and Cl

(D) Cl and Cl

**Answer 55: (C)**

**Question 56:** Which is a wrong statement?

(A) Rate constant  $k$  = Arrhenius constant  $A$  : if  $E_a = 0$

(B)  $e^{-E_a/RT}$  gives the fraction of reactant molecules that are activated at the given temp

(C)  $\ln k$  vs  $\frac{1}{T}$  plot is a straight line

(D) Presence of catalyst will not alter the value of  $E_a$

**Answer 56: (D)**

**Question 57:** 1L of 2 M CH<sub>3</sub>COOH is mixed with 1L of 3M C<sub>2</sub>H<sub>5</sub>OH to form an ester. The rate of the reaction with respect to the initial rate when each solution is diluted with an equal volume of water will be

(A) 0.25 times

- (B) 2 times
- (C) 0.5 times
- (D) 4 times

**Answer 57: (A)**

**Question 58:** Which of the following is an example of homogeneous catalysis?

- (A) Oxidation of  $\text{NH}_3$  in Ostwald's process
- (B) Oxidation of  $\text{SO}_2$  in contact process
- (C) Oxidation of  $\text{SO}_2$  in lead chamber process
- (D) Manufacture of  $\text{NH}_3$  by Haber's process

**Answer 58: (C)**

**Question 59:** Critical Micelle concentration for a soap solution is  $1.5 \times 10^{-4} \text{ mol L}^{-1}$ . Micelle formation is possible only when the concentration of soap solution in  $\text{mol L}^{-1}$  is

- (A)  $2.0 \times 10^{-3}$
- (B)  $4.6 \times 10^{-5}$
- (C)  $7.5 \times 10^{-5}$
- (D)  $1.1 \times 10^{-4}$

**Answer 59: (A)**

**Question 60:** Oxidation state of copper is +1 in

- (A) Malschite
- (B) Cuprite
- (C) Azurite
- (D) Chalcopyrite

**Answer 60: (B)**